Modeling Volatility

References:

Tsay (2010), Ch. 3; Enders (2014), Ch. 3; Hamilton (1994), Ch. 21; Pesaran (2015), Ch. 18

1 Introduction

- Volatility represents the rate at which asset price (or output) moves up and down. It is a measure of uncertainty/risk. In most cases, the higher the volatility, the riskier the asset.
- Stylized characteristics of financial time series (Tsay, 2010, p. 111)
  - There exists *volatility clustering*—volatility tends to be high for certain time periods and low for other periods.
  - Volatility is time-varying, and it evolves over time in a continuous manner (i.e., jumps are rare).
  - Volatility varies within some fixed range and does not diverge to infinity.
  - Volatility seems to react differently to a large price increase and decrease (a.k.a. *leverage effect*, asymmetric volatility).
- Volatility is not directly observable. There are many different ways to measure volatility (such as standard deviation of returns, beta coefficients, option pricing models). We will focus on the econometric analysis of volatility (conditional variance).
- Let y_t denote the series under consideration (e.g., asset returns, inflation rate, exchange rate, output growth) and Ω_t denote the information set at time t. Then the (conditional) volatility of y_t is defined as the conditional variance:

$$h_t = Var\left(y_t|\Omega_{t-1}\right).$$

- However, an ARMA model for $\{y_t\}$ has the property that h_t is independent of t and $\{h_s, s < t\}$.

- More generally, consider the regression model

$$y_t = \boldsymbol{\beta}' \boldsymbol{x}_{t-1} + \varepsilon_t.$$

Under classical assumptions, the disturbances ε_t has a constant variance (*homoskedasticity*) both *unconditionally* and *conditionally*. However, this assumption is invalid in many economic and financial applications. We will discuss conditional heteroskedastic models that are concerned with the evolution of h_t.

2 ARCH Models

- Engle (1982) proposes simultaneously modeling the mean and variance of a series. To illustrate the idea, suppose that

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t. \tag{1}$$

The conditional mean of y_{t+1} is

$$E_t y_{t+1} \equiv E\left(y_{t+1} | \Omega_t\right) \equiv \hat{y}_{t+1|t} = \phi_0 + \phi_1 y_t,$$

and the conditional variance of y_{t+1} is

$$Var\left(y_{t+1} | \Omega_t\right) = E_t\left(y_{t+1} - E_t y_{t+1}\right)^2 = E_t\left[\left(y_{t+1} - \phi_0 - \phi_1 y_t\right)^2\right] = E_t\left(\varepsilon_{t+1}^2\right).$$

Previously we have assumed $E_t\left(\varepsilon_{t+1}^2\right) = \sigma^2$. Now suppose that the conditional variance is time-varying. One simple strategy is to model the conditional variance as an autoregressive process using the *squares* of the estimated residuals from (1), denoted by $\{\hat{\varepsilon}_t\}$:

$$\hat{\varepsilon}_t^2 = \alpha_0 + \alpha_1 \hat{\varepsilon}_{t-1}^2 + \alpha_2 \hat{\varepsilon}_{t-2}^2 + \ldots + \alpha_q \hat{\varepsilon}_{t-q}^2 + v_t, \tag{2}$$

where v_t is a white noise process. Then you can forecast the conditional variance at $t+1$ as

$$E_t\hat{\varepsilon}_{t+1}^2 = \alpha_0 + \alpha_1\hat{\varepsilon}_t^2 + \alpha_2\hat{\varepsilon}_{t-1}^2 + \ldots + \alpha_q\hat{\varepsilon}_{t+1-q}^2.$$

- The residuals in (1) can come from very general ARMA models or standard regression models.

- In practice, the conditional mean and conditional variance of y_t are best estimated jointly by the maximum likelihood method. A better way is to specify ε_t as a multiplicative disturbance, namely,

$$\varepsilon_t = \sqrt{h_t}v_t, \tag{3}$$

where $v_t \sim IID\,(0,1)$, h_t (known as the *volatility* at time t) is related to the past values of ε_t by

$$h_t = \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \ldots + \alpha_q\varepsilon_{t-q}^2,$$

q is some positive integer, $\alpha_0 > 0$ and $\alpha_i \geq 0$, $i = 1, 2, \ldots, q$. $\{\varepsilon_t\}$ is said to follow an *autoregressive conditional heteroskedasticity* process of order q, or ARCH(q) process. Note that ε_t is a function of v_t, v_{t-1}, v_{t-2}, ...

Exercise. Consider the ARCH(1) model:

$$h_t = \alpha_0 + \alpha_1\varepsilon_{t-1}^2,$$

where $\alpha_0 > 0$ and $\alpha_1 \geq 0$. Derive the $t-1$ conditional mean, unconditional mean, $t-1$ conditional variance, unconditional variance, and autocovariance functions of $\{\varepsilon_t\}$.

- In sum, ARCH(1) has the following properties
  - The unconditional and conditional expectations of the error terms are equal to zero: $E(\varepsilon_t) = 0$, $E_{t-1}(\varepsilon_t) = 0$.
  - The unconditional variance is a constant (if $|\alpha_1| < 1$). The conditional variance is time-varying and is an autoregressive process resulting in conditionally heteroskedastic errors: $Var(\varepsilon_t) = \alpha_0/(1-\alpha_1)$, $Var(\varepsilon_t|\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots) = h_t = \alpha_0 + \alpha_1\varepsilon_{t-1}^2$.
  - The $\{\varepsilon_t\}$ sequence is serially uncorrelated: $E(\varepsilon_t\varepsilon_{t-s}) = E_{t-1}(\varepsilon_t\varepsilon_{t-s}) = 0$, for $s > 0$. But $\{\varepsilon_t\}$ are NOT independent since they are related through their second moment.
  - $\alpha_0 > 0$, $0 \leq \alpha_1 < 1$. (Unconditionally the ARCH(1) model is stationary if $|\alpha_1| < 1$, which reduces to the requirement $\alpha_1 < 1$ since α_1 is nonnegative.)

Exercise. Show that the above properties generalizes to ARCH(q) models.

- For ARCH(q) models:

$$h_t = Var(\varepsilon_t|\Omega_{t-1}) = \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \ldots + \alpha_q\varepsilon_{t-q}^2. \tag{4}$$

  Clearly, $E(\varepsilon_t) = E_{t-1}(\varepsilon_t) = 0$. The unconditional variance is

$$Var(\varepsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \ldots - \alpha_q},$$

  if all the roots of $\alpha(z) = 1 - \sum_{i=1}^q \alpha_i z^i = 0$ lie outside the unit circle. Given the nonnegativity constraint, ε_t^2 is stationary if $\sum_{i=1}^q \alpha_i < 1$. Note that the $\{\varepsilon_t\}$ sequence are uncorrelated but not independent.

- How does the properties of ε_t affect $\{y_t\}$? Consider the AR(1)-ARCH(1) model as an illustration:

$$\begin{aligned} y_t &= \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, \\ \varepsilon_t &= \sqrt{h_t} v_t, \quad v_t \sim IID(0,1) \\ h_t &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2. \end{aligned}$$

– The conditional mean and conditional variance are given by

$$E_{t-1}y_t = E\left(\phi_0 + \phi_1 y_{t-1} | y_{t-1}, y_{t-2}, \ldots\right) = \phi_0 + \phi_1 y_{t-1}.$$

$$\begin{aligned} Var\left(y_t | y_{t-1}, y_{t-2}, \ldots\right) &= E_{t-1}\left(y_t - E_{t-1}y_t\right)^2 = E_{t-1}\left(y_t - \phi_0 - \phi_1 y_{t-1}\right)^2 \\ &= E_{t-1}\left(\varepsilon_t^2\right) = h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2. \end{aligned}$$

Since $\alpha_1 \geq 0$, the conditional variance of y_t is positively related to ε_{t-1}^2.

– The unconditional mean is $Ey_t = \phi_0/\left(1-\phi_1\right)$ if $|\phi_1| < 1$, since $E\varepsilon_t = 0$. The unconditional variance of y_t is

$$Var\left(y_t\right) = \sum_{i=0}^{\infty} \phi_1^{2i} Var\left(\varepsilon_{t-i}\right) = \left(\frac{1}{1-\phi_1^2}\right)\left(\frac{\alpha_0}{1-\alpha_1}\right),$$

since $E\left(\varepsilon_t \varepsilon_{t-s}\right) = 0$ for all $s \neq 0$, and $Var\left(\varepsilon_t\right) = \alpha_0/\left(1-\alpha_1\right)$ for all t.

3 GARCH Models

- Bollerslev (1986) extended the ARCH models by allowing the conditional variance to be an ARMA process. Now the error process follows

$$\varepsilon_t = \sqrt{h_t} v_t,$$

where $v_t \sim IID\left(0,1\right)$,

$$h_t = \alpha_0 + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j h_{t-j}, \tag{5}$$

$\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^{\max(p,q)} \left(\alpha_i + \beta_i\right) < 1$.[1] This generalized ARCH model is referred to as the GARCH(p,q) model, which allows for both autoregressive and moving average components in the heteroskedastic conditional variance. The parameters, α_i and β_j, are referred to as ARCH and GARCH parameters, respectively.

[1] It should be understood that $\alpha_i = 0$ for $i > q$, and $\beta_j = 0$ for $j > p$.

- ARCH models can be viewed as special cases of GARCH models.
  - ARCH(1) model is a GARCH(0,1) model, where $p = 0$ and $q = 1$.
  - ARCH(q) model is a GARCH($0, q$) model, where $p = 0$ (or all values of β_i equal zero).
- The main benefit of the GARCH model is that it is more parsimonious than a high-order ARCH model and thus is much easier to identify and estimate.
- Let us understand the key properties of GARCH models by examining the GARCH(1,1) model where

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}. \tag{6}$$

  Define $\eta_t = \varepsilon_t^2 - h_t$, which is the volatility surprise (or the error associated with the volatility forecast). It can be shown that $E(\eta_t) = 0$ and $E(\eta_t \eta_{t-s}) = 0$ for $s \geq 1$, but in general $\{\eta_t\}$ is NOT an i.i.d. sequence. The conditional heteroskedasticity of ε_t is inherited by η_t. Plugging $h_t = \varepsilon_t^2 - \eta_t$ into (6), we obtain

$$\varepsilon_t^2 - \eta_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \left(\varepsilon_{t-1}^2 - \eta_{t-1} \right),$$

  and then

$$\varepsilon_t^2 = \alpha_0 + (\alpha_1 + \beta_1) \varepsilon_{t-1}^2 + \eta_t - \beta_1 \eta_{t-1}. \tag{7}$$

  It can be seen from (7) that ε_t^2 is an ARMA(1,1) process, which is stationary if $|\alpha_1 + \beta_1| < 1$. If $\alpha_1 \geq 0$ and $\beta_1 \geq 0$, then the stationary condition reduces to $\alpha_1 + \beta_1 < 1$. Under the stationary condition,

$$Var(\varepsilon_t) = E\left(\varepsilon_t^2\right) = \frac{\alpha_0}{1 - (\alpha_1 + \beta_1)}.$$

- By the same arguments, it can be shown that the GARCH(p, q) model can be rewritten as

$$\varepsilon_t^2 = \alpha_0 + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) \varepsilon_{t-i}^2 + \eta_t - \sum_{j=1}^{p} \beta_j \eta_{t-j},$$

  which is an ARMA(r, p) process, where $r = \max(p, q)$. Note that it should be understood that $\alpha_i = 0$ for $i > q$, and $\beta_j = 0$ for $j > p$.

- Estimation: Conditional MLE
  - Idea: Consider the ARCH(q) model. The likelihood function is given by

$$f(\varepsilon_1, \ldots, \varepsilon_T|\boldsymbol{\alpha}) = f(\varepsilon_T|\Omega_{T-1}) f(\varepsilon_{T-1}|\Omega_{T-2}) \times \ldots \times f(\varepsilon_{q+1}|\Omega_q) f(\varepsilon_1, \ldots, \varepsilon_q|\boldsymbol{\alpha}),$$

    where $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \ldots, \alpha_q)'$. The exact form of $f(\varepsilon_1, \ldots, \varepsilon_q|\boldsymbol{\alpha})$ is complicated. If it is dropped, we obtain the conditional likelihood function, which is given by

$$f(\varepsilon_{q+1}, \ldots, \varepsilon_{q+T}|\boldsymbol{\alpha}, \varepsilon_1, \ldots, \varepsilon_q) = \prod_{q+1}^{T} \frac{1}{\sqrt{2\pi h_t}} \exp\left(-\frac{\varepsilon_t^2}{2h_t}\right), \tag{8}$$

    if ε_t follows a normal distribution.
  - In some applications, it is more appropriate to assume that ε_t follows a heavy-tailed distribution such as t distribution. Then the probability density function in (8) need to be replaced, and the estimates are referred to as the conditional MLEs under t distribution.
- Forecasting: Forecasts of a GARCH model can be obtained by similar methods as those of an ARMA model.

Exercise. Consider the ARCH(q) model given by(4). Derive the h-step ahead forecasts at the forecast origin t, for $h \geq 1$.

Solution. Forecasts of ARCH models can be obtained recursively as those of AR models. The 1-step-ahead forecast ($h = 1$) for h_{t+1} is

$$\hat{h}_{t+1|t} = E(h_{t+1}|\Omega_t) = \alpha_0 + \alpha_1 \varepsilon_t^2 + \alpha_2 \varepsilon_{t-1}^2 + \ldots + \alpha_q \varepsilon_{t+1-q}^2,$$

where $\varepsilon_t, \varepsilon_{t-1}, \ldots, \varepsilon_{t+1-q}$ are known at time t. The 2-step-ahead forecast is

$$\begin{aligned}\hat{h}_{t+2|t} &= E(h_{t+2}|\Omega_t) = \alpha_0 + \alpha_1 E\left(\varepsilon_{t+1}^2|\Omega_t\right) + \alpha_2 \varepsilon_t^2 + \ldots + \alpha_q \varepsilon_{t+2-q}^2 \\ &= \alpha_0 + \alpha_1 \hat{h}_{t+1|t} + \alpha_2 \varepsilon_t^2 + \ldots + \alpha_q \varepsilon_{t+2-q}^2.\end{aligned}$$

The h-step-ahead forecast is

$$\hat{h}_{t+h|t} = \alpha_0 + \sum_{i=1}^{q} \alpha_i \hat{h}_{t+h-i|t},$$

where $\hat{h}_{t+h-i|t} = h_{t+h-i}$ if $h - i \leq 0$.

Exercise. Consider the GARCH(1,1) model given by (6). Derive the h-step ahead forecasts at the forecast origin t, for $h \geq 1$.

Solution. For $h = 1$, we have

$$h_{t+1} = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 h_t,$$

where ε_t and h_t are known at time t. Therefore, the 1-step-ahead forecast is

$$\hat{h}_{t+1|t} = E\left(h_{t+1}|\Omega_t\right) = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 h_t.$$

For $h = 2$, the volatility equation is

$$h_{t+2} = \alpha_0 + \alpha_1 \varepsilon_{t+1}^2 + \beta_1 h_{t+1},$$

and then the 2-step-ahead forecast is given by

$$\hat{h}_{t+2|t} = E\left(h_{t+2}|\Omega_t\right) = \alpha_0 + \alpha_1 E\left(\varepsilon_{t+1}^2|\Omega_t\right) + \beta_1 E\left(h_{t+1}|\Omega_t\right) = \alpha_0 + \left(\alpha_1 + \beta_1\right)\hat{h}_{t+1|t}.$$

In general,

$$\hat{h}_{t+h|t} = \alpha_0 + \left(\alpha_1 + \beta_1\right)\hat{h}_{t+h-1|t}, \quad h > 1. \tag{9}$$

Remark. This result is the same as that of an ARMA(1,1) model with AR polynomial $1 - \left(\alpha_1 + \beta_1\right)L$. By repeated substitution in (9), the h-step forecast can be written as

$$\hat{h}_{t+h|t} = \frac{\alpha_0\left[1 - \left(\alpha_1 + \beta_1\right)^{h-1}\right]}{1 - \alpha_1 - \beta_1} + \left(\alpha_1 + \beta_1\right)^{h-1}\hat{h}_{t+1|t},$$

and hence

$$\hat{h}_{t+h|t} \to \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \text{as } h \to \infty,$$

if $\alpha_1+\beta_1 < 1$. That is, the multistep-ahead volatility forecasts of a GARCH(1,1) model converge to the unconditional variance of ε_t (provided that $Var(\varepsilon_t)$ exists), as the forecast horizon tends to infinity.

- Strengths and weaknesses of GARCH models
  - GARCH models describe volatility evolution using simple parametric functions.
  - They are able to generate volatility clustering: A large ε_{t-1}^2 or h_{t-1} leads to a large h_t, meaning that high ε_{t-1}^2 tends to be followed by high ε_t^2.
  - They can generate tail distributions that are heavier than normal distributions if the parameters satisfy additional constraints. For example, for the GARCH(1,1) model, it can be shown that the unconditional kurtosis of ε_t is

$$\frac{E(\varepsilon_t^4)}{[E(\varepsilon_t^2)]^2} = \frac{3\left[1-(\alpha_1+\beta_1)^2\right]}{1-(\alpha_1+\beta_1)^2-2\alpha_1^2} > 3$$

    if $1-(\alpha_1+\beta_1)^2-2\alpha_1^2 > 0$.
  - The models assume that positive and negative shocks have the same effects on volatility.
  - The models do not explain the source of variations of a series.

4 Building a GARCH Model

- Step 1: Modeling the mean of a series and testing for ARCH effects.
  - H_0 : no ARCH effects vs. H_1 : ARCH effects
  - Ljung-Box Q-statistics of squared residuals (McLeod and Li, 1983)
  - Lagrange multiplier test (Engel, 1982), which involves the following two steps: First, estimate the mean equation (ARMA or other general regression models). Denote the residuals by $\hat{\varepsilon}_t$. Second, regress $\hat{\varepsilon}_t^2$ on a constant and q of its own lagged values:

$$\hat{\varepsilon}_t^2 = \alpha_0 + \alpha_1\hat{\varepsilon}_{t-1}^2 + \ldots + \alpha_q\hat{\varepsilon}_{t-q}^2 + \text{error}. \tag{10}$$

A test of ARCH(q) effect can be carried out by testing the null of $\alpha_1 = \alpha_2 = \ldots = \alpha_q = 0$. The Lagrange test statistic$= TR^2 \sim \chi^2(q)$, where T is the sample size, and R^2 is the usual R-squared of the regression given by (10).

  - Testing for GARCH effects can be achieved using the same tests for the ARCH effects, since GARCH models can be approximated by ARCH representations.

- Step 2: Order determination: Examine PACF of squared residuals, or maybe simply try some reasonable order in practice.
  - ARCH has poor performance, yet it started the whole enterprise.
  - GARCH(1,1) is the most widely used GARCH process.
- Step 3: Estimation: Conditional MLE
- Step 4: Diagnostic checking: The standardized residuals, $\tilde{\varepsilon}_t = \varepsilon_t/\sqrt{h_t}$, form a sequence of i.i.d. random variables if the fitted GARCH model is adequate.
  - Apply the Ljung-Box test to $\tilde{\varepsilon}_t$ so check the adequacy of the mean equation
  - Apply the Ljung-Box test to $\tilde{\varepsilon}_t^2$ to check the adequacy of the volatility equation
  - Check validity of the distribution assumption: skewness, kurtosis, quantile-to-quantile plot of $\{\tilde{\varepsilon}_t\}$
- R commands: `garchFit` in the `fGarch` package; `ugarchfit` in the `rugarch` package